

Test plan

All components are thoroughly tested for their core functionalities during development, using manual tests or unit tests.

The software components are being tested with automatic tests that are executed at the end of the CI/CD pipeline, while the hardware components and the controller were tested manually in the environment where the robot is used.

Manual testing has proven to be more effective for the scale of this project, particularly given the tight schedule and the need for quick identification of issues. Components that interact with the physical environment, such as motors and limit switches, must be tested manually under real-world conditions. This is because unit tests alone cannot uncover issues arising from the interaction between the software and the hardware. Such issues include gear backlash, slippage, or inaccuracies in motor positioning, all of which require direct observation.

The piece position detection is tested through a manually generated image gallery of different chess positions under various light conditions.

The Frontend was tested with the integrated testing tool, which emulates a browser and verifies the DOM tree while executing the functions and checking.

GameManager and ChessEngine are intertwined; the `deduce_move` method, as the most important method of this class, was tested in a unit test, and the rest was tested manually using the frontend.

Integration tests are conducted at the end to ensure that the combination of all components works well together and to identify any last problems that may arise when using all elements together for a functioning robot.